

Treasure Coast Global Property Solutions

Platform Architecture + Product Requirements Document (PRD)

Status

Draft v1 for planning and legal review before development

1. Product Summary

Treasure Coast Global Property Solutions (TCGPS) is a web-based private real estate investment platform that will begin as an LLC-based investment vehicle and may evolve toward REIT status later. The platform will allow users to create accounts, complete onboarding, sign required documents, fund investments, view their holdings, receive quarterly distributions, and optionally reinvest.

The platform will also include a restricted internal admin environment for board members and property managers to manage properties, investor records, reports, onboarding status, distributions, and communications.

This product must be designed to support: - investor trust - secure onboarding - legal/compliance adaptability - role-based data protection - accounting coordination with an external QuickBooks CPA - future evolution as legal structure becomes more formalized

2. Core Product Goals

Business goals

- Create a polished public-facing investment platform that builds confidence with prospective investors.
- Enable efficient investor onboarding and document collection.
- Provide a secure investor portal for holdings, documents, reports, and distributions.
- Provide a hidden internal operations portal for board members and property managers.
- Preserve clean, exportable investment and transaction data for CPA/accounting use.
- Support future legal and operational expansion without rebuilding the platform from scratch.

User goals

Prospective investor

- Learn about the company and properties

- Create a free account
- Complete onboarding and legal documents
- Invest money securely
- Track investment status over time

Active investor

- View portfolio balance / estimated value
- View only the properties they are invested in
- Access signed documents
- See quarterly distributions and reinvestment options
- Contact support

Board member / admin / property manager

- Manage investor onboarding
 - Manage properties and related reports
 - Control what data is visible to which users
 - Upload documents and investor communications
 - Track transactions and distributions
 - Export structured data for finance and compliance support
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3. Recommended Platform Architecture

3.1 High-level architecture

Public site

- Marketing pages
- Company overview
- Property showcase (public-safe information only)
- FAQ / education
- Contact / lead capture
- Investor sign-up

Authenticated investor application

- Investor dashboard
- Portfolio / balances
- Property views for invested properties only
- Distribution history
- Reinvestment controls
- Documents vault
- Support messaging
- Profile / security settings

Internal operations application

- Hidden admin portal
- Investor management
- Onboarding review
- Property management
- Report uploads
- Distribution management
- Document administration
- Messaging / notices
- Audit and activity logs

Shared backend

- Auth
 - Database
 - File storage
 - Business logic / workflows
 - Email + notifications
 - Payment integrations
 - Document-signing integration
 - Reporting and exports
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3.2 Recommended stack

Frontend

- Next.js preferred for app + marketing site in one codebase
- React component architecture
- Tailwind CSS or modular styling system

Backend / data

- Supabase PostgreSQL
- Supabase Auth
- Supabase Storage
- Supabase Edge Functions for workflows
- Row Level Security for permissioning

Integrations

- DocuSign for legal document signing
- Plaid for bank account linking and verification
- ACH processor to actually move money (to be selected after legal/ops review)
- QuickBooks handled outside the product by CPA, with exports/imports from the platform
- Email provider such as Resend / SendGrid / Postmark

Infrastructure

- Separate environments: dev / staging / production
 - Separate file buckets for public assets and private investor/admin documents
 - Audit logging on sensitive actions
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4. Hidden Admin Portal Strategy

The admin portal does NOT need to be a separate database or completely separate product. It should be a restricted part of the same platform with strong role-based access controls.

Recommendation

Use: - one codebase - one backend - one database - separate route groups / layouts - role-based access gating

Example route structure

- `/` public marketing pages
- `/app` investor application
- `/ops` internal operations portal

How to keep investors from knowing it exists

- Do not place links to admin routes anywhere in investor navigation.
- Block access to `/ops` at the middleware/auth level unless role is admin, board_member, or property_manager.
- Return `404 Not Found` for unauthorized users instead of a visible permission error.
- Exclude admin assets and menu structures from investor bundles when possible.
- Use a separate subdomain if desired for extra separation, e.g. `ops.domain.com`, while still pointing to the same backend.

Best recommendation

Use a separate subdomain for internal operations: - `www.domain.com` → public + investor app - `ops.domain.com` → admin/property manager portal

Both can use the same backend and database, but the separation helps with: - cleaner UX - reduced investor visibility - easier internal branding and navigation - better security posture

5. Role Model

Roles

- visitor
- prospective_investor
- investor
- property_manager
- board_member
- admin
- support

Role behavior

Visitor

- browse public site
- view public-safe property summaries
- register account

Prospective investor

- complete onboarding
- sign docs
- submit required info
- cannot invest until approved

Investor

- view their account
- view transactions
- view signed docs
- view distributions
- reinvest
- view only properties they are invested in
- message support

Property manager

- manage property records
- upload reports and metrics
- publish investor-facing property information
- cannot necessarily access full investor admin tools unless granted

Board member

- broader visibility into investor activity and property performance

- view distributions and governance-level reporting

Admin

- full platform control
 - user management
 - onboarding approvals
 - document controls
 - permissions
 - reporting exports
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6. Data Visibility Rules

Public users can see

- company overview
- educational information
- selected property summaries
- property type, market, imagery, descriptions
- never exact addresses if that is the chosen business rule

Logged-in investors can see

- their own account data
- their own documents
- their own transactions
- their own distributions
- only the properties they are invested in

Sensitive property metrics must be hidden unless investor is invested in that property

The following must only be visible when a user has an active investment allocation in that property: - current estimated value - occupancy rate - revenue - expenses - quarterly income - investor return %

Implementation note

Use per-property entitlement checks driven by an `investments` table. If no active relation exists between user and property, the API should not return sensitive property fields.

7. Full Feature Requirements

7.1 Public Website

Required

- homepage
- about / company story
- how it works
- property showcase
- FAQ
- contact form
- account creation / investor signup CTA

Should plan for

- educational content pages
- legal disclaimers page
- news / updates
- event pages

Could implement later

- blog / insights
 - referral / ambassador pages
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7.2 Account Creation and Authentication

Required

- email/password account creation
- email verification
- secure login
- password reset

Should plan for

- 2FA / MFA for investors and mandatory MFA for admins
- session management
- device/session alerts for admins

Could implement later

- passkeys
- magic links

Implementation: - Supabase Auth - middleware-based route protection - optional TOTP MFA depending on product and legal preference

7.3 Investor Onboarding

Required

- create free account
- complete onboarding questionnaire
- sign required legal documents
- wait for approval or automated qualification step before investing

Must be confirmed with lawyer before final design

- investor eligibility requirements
- accredited / non-accredited rules
- suitability requirements
- KYC/AML needs
- rescission / consent requirements
- withdrawal / redemption language

Should plan for

- onboarding status tracker
- admin review queue
- resend document packet
- reminder emails for incomplete onboarding

Preferred implementation

- DocuSign integration
- webhook updates onboarding status when docs are completed

Alternative (not preferred now) - internal signature capture tool

Recommendation: Use DocuSign first. Easier legally, easier audit trail, easier trust.

7.4 Investor Dashboard

Required

- total balance / contributed amount
- estimated investment value
- quarterly distribution summary
- reinvestment status

- recent account activity
- document alerts / support alerts

Should plan for

- charts over time
- portfolio allocation by property
- pending actions widget

Could implement later

- benchmark comparison
- annual performance view

Implementation: - server-rendered dashboard using protected queries - value calculations from investment
+ distribution records + property valuation snapshots

7.5 Investments Module

Required

- list of active investments
- amount invested
- investment date
- distribution history
- reinvestment history
- redemption/withdrawal history if allowed

Should plan for

- downloadable statements
- tax-lot style records if needed later

Must be clarified legally/operationally

- whether withdrawals are allowed
 - lock-up periods
 - redemption windows
 - how partial exits are handled
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7.6 Property Views

Required

- investor can view only invested properties
- non-sensitive public version for general site visitors

- no exact property addresses shown to investors if that rule remains
- reports available for invested properties only

Investor property detail may include

- property name
- asset class
- city / market
- acquisition date
- summary description
- report downloads
- selected images / updates

Sensitive metrics only for invested users

- current estimated value
- occupancy rate
- revenue
- expenses
- quarterly income
- investor return %

Admin requirements

- upload property files
 - tag which investors / groups can access each report if needed
 - maintain valuation snapshots over time
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7.7 Reinvestment and Distributions

Required

- quarterly distribution record
- show paid / pending / reinvested status
- allow investor to choose reinvestment option if permitted

Should plan for

- global reinvestment election setting
- one-time reinvestment action
- admin override with audit log

Must be clarified

- whether reinvestment is account-level or property-level
- how reinvestment changes ownership records
- timing rules around quarter close

Implementation: - distributions ledger table - election table or reinvestment preference field - admin distribution run workflow

7.8 Payments and Transfers

Required

- connect bank account
- fund investments
- receive payouts
- track all transfer statuses

Important note

Plaid handles account linking/verification, but another processor is needed to actually move ACH funds.

Should plan for

- pending / completed / failed transfer states
- manual review flags
- reconciliation exports for CPA

Must be clarified

- which ACH processor is chosen
 - who can initiate pull / push transactions
 - failed payout handling
 - refund / reversal handling
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7.9 Documents Vault

Required

- investors can view and download the documents they signed
- admins can upload and manage documents
- private documents segmented by user
- property reports stored privately

Should plan for

- versioning
- expiration reminders
- admin ability to request re-signing when docs are updated

Implementation: - Supabase Storage private buckets - metadata tables mapping document ownership and visibility - DocuSign envelope/document IDs stored for traceability

7.10 Support Messaging

Required

- investor can message support
- admin/support users can respond
- status tracking (open, in progress, closed)

Should plan for

- email notifications
- internal notes visible only to admins
- attachment support

Could implement later

- live chat
 - knowledge base integration
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7.11 Admin / Property Manager Portal

Required admin functions

- manage users and roles
- view onboarding queue
- approve / reject / flag onboarding
- manage properties
- upload reports
- manage investor-property relationships
- manage distribution records
- manage document visibility
- respond to support tickets
- create announcements / notices

Recommended additional admin features

- dashboard with KPIs
- property valuation snapshot entry
- report publishing controls
- audit log viewer
- export center for CPA / legal / board
- internal notes on investor accounts

- distribution run preparation page
- failed payment review queue
- communications center
- investor segmentation / tagging
- report acknowledgment tracking

Board-specific capabilities

- higher-level portfolio summary
- investor totals
- property performance summaries
- governance-ready reporting exports

Property manager-specific capabilities

- update non-financial property details
- upload property photos and updates
- upload quarterly operating reports
- update occupancy / unit mix / status
- limited visibility into investor records based on permissions

7.12 Reporting and Exports

Required

- export transactions
- export investor roster
- export distribution history
- export property report metadata
- export signed document records

Should plan for

- accountant-specific export templates
- board reporting packs
- quarterly performance summaries

Implementation: - CSV / Excel export endpoints - filtered reporting UI - secure audit logging of exports

8. Operational Workflows

8.1 New investor workflow

1. user visits website
2. creates free account

3. verifies email
4. completes onboarding form
5. receives DocuSign packet
6. signs documents
7. admin or automated workflow marks onboarding complete
8. bank account link step begins
9. funding enabled
10. investment recorded
11. investor gains access to invested-property details only

8.2 Property update workflow

1. property manager logs in to ops portal
2. updates property summary and uploads report
3. system saves metadata and access level
4. eligible investors see update in their portal
5. optional investor notification sent

8.3 Quarterly distribution workflow

1. admin prepares quarter distribution data
2. review against accounting inputs / QuickBooks support files
3. distribution records created
4. payout or reinvestment action executed
5. investor dashboard updated
6. statement/export generated

9. Security Requirements

Required

- role-based access control
- row-level security
- encrypted document storage
- audit logs for sensitive actions
- MFA mandatory for admin users
- secure webhook verification for DocuSign/payment events
- environment secret management
- rate limiting for auth and sensitive actions

Should plan for

- MFA for investors
- IP / device monitoring for admin access
- signed URLs for private document access

- inactivity timeouts for admin portal

Could implement later

- anomaly detection
 - admin approval for high-risk actions
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10. Out of Scope for Initial Build

- full internal accounting system replacing QuickBooks
 - public marketplace / trading system for investor shares
 - complex tax engine
 - automated compliance determinations without legal approval
 - mobile app at launch unless required later
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11. Major Open Questions Before Final System Design

Legal / lawyer meeting

- who can legally invest?
- is this accredited-only, mixed, or otherwise restricted?
- what exact onboarding docs are required?
- do we need KYC/AML?
- can users freely withdraw, or are there lockups?
- what is the legal structure of reinvestment?
- what disclosures must exist before funding is enabled?

Business / operations

- are investments tied to specific properties, pooled funds, or both?
- how is estimated value calculated?
- how often is valuation updated?
- how are distributions approved and reconciled?
- what exact redemption process will exist?
- what metrics should property managers be allowed to edit directly?

Finance / accounting

- what data export format does the CPA need?
- how often will the platform reconcile with QuickBooks?

- what transaction fields must be captured for audit and accounting use?
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12. Suggested Development Phases

Phase 0 – Planning / legal alignment

- lawyer meeting
- workflow confirmation
- data model finalization
- PRD approval
- security model approval

Phase 1 – Foundation

- public site
- auth
- role system
- investor dashboard shell
- ops portal shell
- properties module base
- documents module base

Phase 2 – Onboarding + documents

- onboarding forms
- DocuSign integration
- approval workflow
- private document vault

Phase 3 – Investments + property entitlements

- investment records
- property visibility rules
- sensitive metrics gating
- reports upload + access

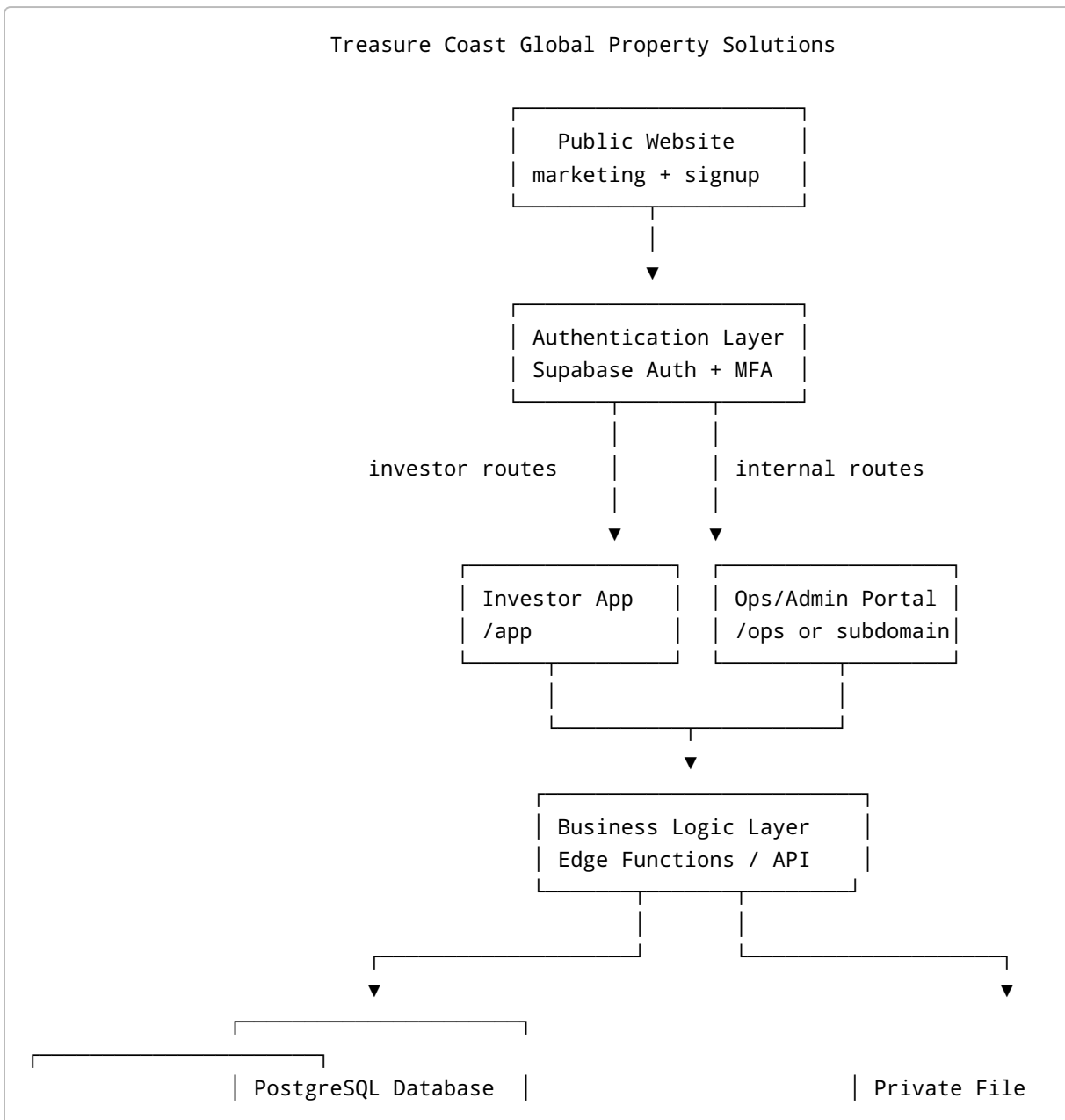
Phase 4 – Money movement + distributions

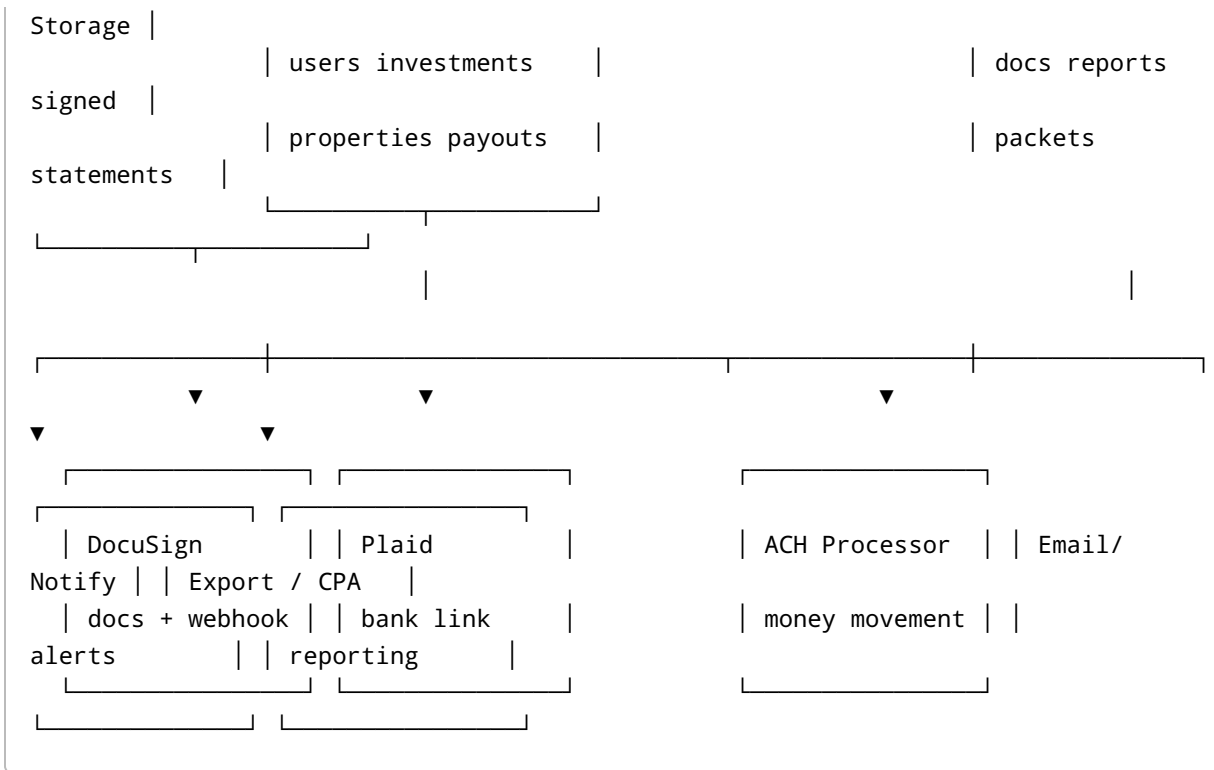
- bank linking
- ACH processor integration
- transaction ledger
- quarterly distribution workflows
- reinvestment options

Phase 5 – Admin maturity

- exports
- analytics
- board reporting
- audit views
- support messaging improvements

13. Architecture Diagram (text version)





14. Recommendation Summary

Recommended architectural direction

- one backend
- one database
- one codebase where practical
- separate investor and ops experiences
- separate subdomain for ops strongly recommended
- DocuSign preferred over custom signing for launch
- QuickBooks remains external
- use role-based permissions and entitlement checks for all sensitive property/investor data

Immediate next deliverables

1. lawyer meeting agenda
2. database schema draft
3. user flow maps
4. route/access matrix
5. onboarding workflow specification